# Enterprise Database Infrastructure Administration Lab

Design, Deployment, Security Hardening, Backup, Replication and Monitoring of PostgreSQL Infrastructure on Linux

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## SUMMARY

# PROJECT DESIGN & ARCHITECTURE

**Enterprise Database Infrastructure Administration Lab**

Design, Deployment, Security Hardening, Backup, Replication and Monitoring of PostgreSQL Infrastructure on Linux

Project Type : Enterprise Infrastructure Project

Environment : VMware Workstation Pro

Platform : Linux

Primary Database : PostgreSQL

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# I. Introduction

## 1. Background

Modern organizations rely heavily on database systems to store, process, and secure critical business information. Database administrators and system administrators are responsible for ensuring that databases remain available, secure, performant, and recoverable in the event of failures.

As organizations increasingly adopt Linux-based infrastructures and open-source technologies, PostgreSQL has become one of the most widely used enterprise-grade relational database management systems due to its reliability, extensibility, security features, and high-performance capabilities.

This project aims to simulate a real-world enterprise database environment by designing, deploying, securing, monitoring, and maintaining a PostgreSQL infrastructure within a virtualized environment.

The project follows industry best practices and focuses on the operational responsibilities typically performed by System Administrators, Database Administrators (DBAs), Infrastructure Engineers, and DevOps professionals.

## 2. Problem Statement

Organizations face several challenges when managing databases :

- Unauthorized access attempts
- Data corruption or accidental deletion
- Service outages
- Storage failures
- Performance degradation
- Backup failures
- Replication issues
- Lack of monitoring and visibility

A properly designed database infrastructure must mitigate these risks while ensuring business continuity and data integrity.

## 3. Project Purpose

The purpose of this project is to build a fully functional enterprise database administration environment capable of supporting production-like operations while providing hands-on experience in :

- Database deployment
- Database administration
- Security hardening
- Backup and recovery
- Replication
- Monitoring
- Troubleshooting
- Automation

# II. Project Objectives

## 1. General Objectives

Design and deploy a secure, resilient, monitored, and manageable PostgreSQL infrastructure in a VMware Workstation environment using Linux servers.

## 2. Specific Objectives

The project will enable the Following :

### i. Infrastructure Administration

- Deploy multiple Linux servers
- Configure static network addressing
- Manage system services
- Configure remote administration

### ii. Database Administration

- Install PostgreSQL
- Create databases
- Manage users and roles
- Configure authentication
- Manage schemas and privileges

### iii. Security

- Harden Linux servers
- Secure database access
- Restrict network exposure
- Implement auditing and logging

### iv. Data Protection

- Configure backups
- Implement recovery procedures
- Validate restoration processes

### v. High Availability

- Deploy PostgreSQL replication
- Test synchronization
- Simulate failover scenarios

### vi. Monitoring

- Monitor infrastructure health
- Monitor database performance
- Collect metrics
- Visualize dashboards

### vii. Operations

- Troubleshoot incidents
- Analyze logs

# III. Project Scope

## 1. Included

The project includes :

### i. Infrastructure

- Linux Server Deployment
- Network Configuration
- SSH Administration

### ii. PostgreSQL Administration

- Installation
- Configuration
- User Management
- Security
- Maintenance

### iii. Data Protection

- Backup
- Restore
- Recovery Testing

### iv. Replication

- Primary Server
- Replica Server
- Synchronization Verification

### v. Monitoring

- Server Monitoring
- PostgreSQL Monitoring
- Alerting

### vi. Automation

- Backup Automation
- Maintenance Tasks
- Scheduled Operations

### vii. Incident Response

- Failure Simulation
- Troubleshooting
- Service Recovery

## 2. Excluded

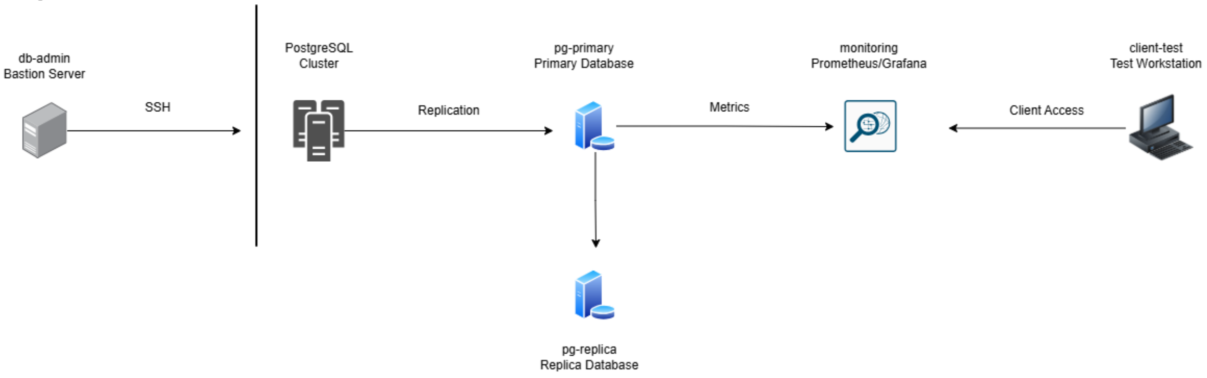
The following items are outside the scope of this project :

- Cloud Deployment
- Kubernetes Integration
- Multi-site Disaster Recovery
- Production Internet Exposure
- Application Development

These may be considered future enhancements.

# IV. Project Architecture

## High-Level Architecture




# V. Virtual Machine Inventory

## 1. VM1 – Administration Server

| Parameter | Value |
|---|---|
| **Hostname** | db-admin |
| **OS** | Ubuntu Server 24.04 LTS |
| **Role** | Administration |
| **vCPU** | 2 |
| **RAM** | 2 GB |
| **Disk** | 20 GB |

**Responsibilities**

- SSH Administration
- Maintenance Operations
- Script Execution
- Future Ansible Controller

## 2. VM2 – PostgreSQL Primary

| Parameter | Value |
|---|---|
| **Hostname** | pg-primary |
| **OS** | Ubuntu Server 24.04 LTS |
| **Role** | Production Database |
| **vCPU** | 2 |
| **RAM** | 4 GB |
| **Disk** | 40 GB |

**Responsibilities**

- Main Database Server
- Transaction Processing
- Backup Source
- Replication Source

## 3. VM3 – PostgreSQL Replica

| Parameter | Value |
|---|---|
| **Hostname** | pg-replica |

| | |
|---|---|
| **OS** | Ubuntu Server 24.04 LTS |
| **Role** | Standby Database |
| **vCPU** | 2 |
| **RAM** | 4 GB |
| **Disk** | 40 GB |

**Responsibilities**

- Replication Target
- Read Replica
- Failover Testing

### 4. VM4 - Monitoring Server

| **Parameter** | **Value** |
|---|---|
| **Hostname** | monitoring |
| **OS** | Ubuntu Server 24.04 LTS |
| **Role** | Monitoring & Backup |
| **vCPU** | 2 |
| **RAM** | 4 GB |
| **Disk** | 40 GB |

**Responsibilities**

- Metrics Collection
- Dashboard Visualization
- Backup Storage
- Alerting

### 5. VM5 – Client Workstation

| **Parameter** | **Value** |
|---|---|
| **Hostname** | client-test |
| **OS** | Ubuntu Server 24.04 LTS |
| **Role** | Database Client |
| **vCPU** | 1 |
| **RAM** | 2 GB |
| **Disk** | 20 GB |

**Responsibilities**

- Database Connectivity Testing
- Query Execution
- User Validation

## VI. Network Design

### 1. Network Information

| **Parameter** | **Value** |
|---|---|
| **Network Type** | VMware Custom Network |
| **Subnet** | 192.168.10.0/24 |
| **Gateway** | 192.168.10.2 |

| | |
|---|---|
| **DNS** | 8.8.8.8 |

### 2. IP Addressing Plan

| Hostname | IP Address |
|---|---|
| **db-admin** | 192.168.10.10 |
| **pg-primary** | 192.168.10.20 |
| **pg-replica** | 192.168.10.21 |
| **monitoring** | 192.168.10.30 |
| **client-test** | 192.168.10.40 |

## VII. Technology Stack

- **Operating System**
  - Ubuntu Server 24.04 LTS
- **Database Platform**
  - PostgreSQL
- **Monitoring**
  - Prometheus
  - Grafana
- **Administration**
  - OpenSSH
  - Bash
  - Cron
- **Automation**
  - Bash Scripting
  - Ansible (future enhancement)

## VIII. Success Criteria

The project will be considered successful if :

- PostgreSQL services operate normally.
- Remote administration functions correctly.
- Database replication is operational.
- Backups are successfully created and restored.
- Monitoring dashboards display real-time metrics.
- Security controls are implemented and validated.
- Incident simulations are completed successfully.
- Complete technical documentation is produced.